Interpretable Machine Learning

Shapley Values for Local Explanations

Learning goals

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- See model predictions as a cooperative game
- Transfer the Shapley value concept from game theory to machine learning

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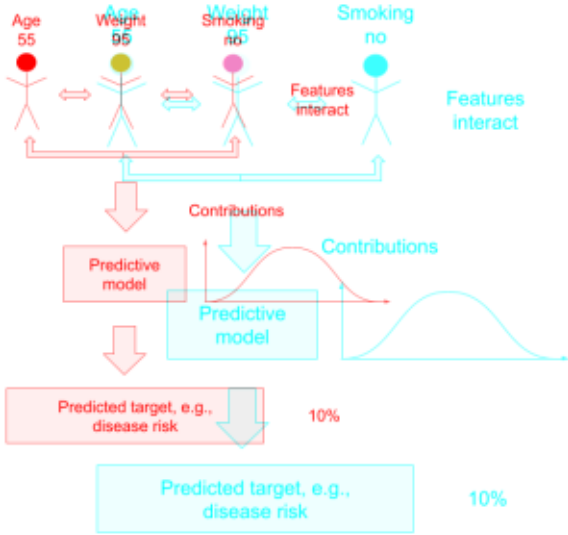
FROM GAME THEORY TO MACHINE LEARNING



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FROM GAME THEORY TO MACHINE LEARNING

- Game: Make prediction $\hat{f}(x_1, x_2, \ldots, x_p)$ for a single observation $\mathbf{x}$

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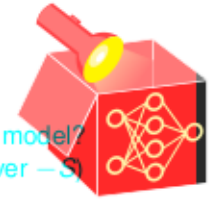
FROM GAME THEORY TO MACHINE LEARNING



- Game: Make prediction $\hat{f}(x_1, x_2, \ldots, x_p)$ for a single observation $\mathbf{x}$
- Players: Features $x_j, j \in \{1, 2, \ldots, p\}$, which cooperate to produce a prediction
  $\rightsquigarrow$ How can we make a prediction with a subset of features without changing the model?
  $\rightsquigarrow$ PD function: $\hat{f}_S(\mathbf{x}_S) := \int_{X_{-S}} \hat{f}(\mathbf{x}_S, X_{-S}) d\mathbb{P}_{X_{-S}}$ ("removing" by marginalizing over $-S$)

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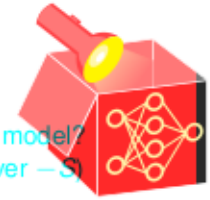
FROM GAME THEORY TO MACHINE LEARNING



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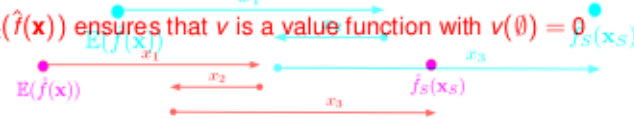
- Value function / payout of coalition $S \subseteq P$ for observation $\mathbf{x}$:
  - $v(S) = \hat{f}_S(\mathbf{x}_S) - \mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$, where $\hat{f}_S : \mathcal{X}_S \mapsto \mathcal{Y}$
  - $\rightsquigarrow$ subtraction of $\mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ ensures that v is a value function with $v(\emptyset) = 0$



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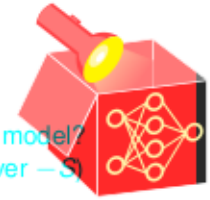

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- Marginal contribution: $v(S \cup \{j\}) - v(S) = \hat{f}_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) - \hat{f}_S(\mathbf{x}_S)$
  - $\rightsquigarrow$ $\mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ cancels out due to the subtraction of value functions

SHAPLEY VALUE - DEFINITION

Shapley (1953) | Strumbelj et al. (2014)

Shapley value ϕ_j of feature j for observation $\mathbf{x}$ via **order definition**:

$$\phi_j(\mathbf{x}) = \frac{1}{|P|!} \sum_{\tau \in \Pi} \underbrace{\hat{f}_{S_j^\tau \cup \{j\}}(\mathbf{x}_{S_j^\tau \cup \{j\}}) - \hat{f}_{S_j^\tau}(\mathbf{x}_{S_j^\tau})}_{\text{marginal contribution of feature } j} - \hat{f}_{S_j^\tau}(\mathbf{x}_{S_j^\tau})$$

- Interpretation: Feature x_j contributed ϕ_j to difference between $\hat{f}(\mathbf{x})$ and average prediction
  $\rightsquigarrow$ Note: Marginal contributions and Shapley values can be negative
- For exact computation of $\phi_j(\mathbf{x})$, the PD function $f_S(\mathbf{x}_S) = \frac{1}{n}\sum_{i=1}^{n} \hat{f}(\mathbf{x}_S, \mathbf{x}_{-S}^{(i)})$ for any set of features S can be used which yields

$$\phi_j(\mathbf{x}) = \frac{1}{|P|! \cdot n} \sum_{\tau \in \Pi} \sum_{i=1}^{n} \hat{f}(\mathbf{x}_{S_j^\tau \cup \{j\}}, \mathbf{x}_{-\{S_j^\tau \cup \{j\}\}}^{(i)}) - \hat{f}(\mathbf{x}_{S_j^\tau}, \mathbf{x}_{-S_j^\tau}^{(i)})$$

$\rightsquigarrow$ Note: $\hat{f}_S$ marginalizes over all other features $-S$ using all observations $i = 1, \ldots, n$

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ESTIMATION: A PRACTICAL PROBLEM

- Exact Shapley value computation is problematic for high-dimensional feature spaces
  For 10 features, there are already $|P|! = 10! \approx 3.6$ million possible orders of features
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Interpretable Machine Learning – 4 / 9

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APPROXIMATION ALGORITHM

Strumbelj et al. (2014)

Estimation of ϕ_j for observation x of model $\hat{f}$ fitted on data D using sample size M:

1. For $m = 1, 1, \ldots, M$ **do**:

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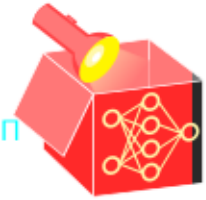

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   5. Compute difference $\phi_j^m = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$
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2. Compute Shapley value $\phi_j = \frac{1}{M} \sum_{m=1}^{M} \phi_j^m$ over M iters.

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SHAPLEY VALUE APPROXIMATION - ILLUSTRATION

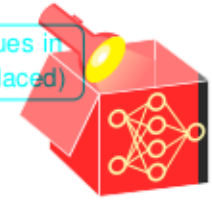
Definition

$\mathbf{x}$: obs. of interest

$\mathbf{x}$ with feature values in S_m (other are replaced)

$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})\right]$$

$\mathbf{x}$ with feature values in $S_m \cup \{j\}$



	Temperature	Humidity	Windspeed	Year
x	10.66	56	11	2012
x_{+j}	10.66	56	11	2012
x_{-j}	10.66	56	random	2012
			j	j

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SHAPLEY VALUE APPROXIMATION - ILLUSTRATION

Definition

Contribution of feature j to coalition S_m

$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\Big[\underbrace{\hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})}_{:=\Delta(j,S_m)}\Big] = \frac{1}{M}\sum_{m=1}^{M}\Big[\underbrace{\hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})}_{:=\Delta(j,S_m)}\Big]$$

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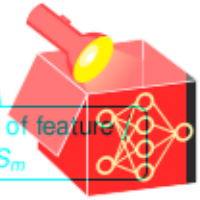

- $\Delta(j, S_m) = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$ is the marginal contribution of feature j to coalition S_m
- Here: Feature *year* contributes +700 bike rentals if it joins coalition $S_m = \{temp, hum\}$

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	Temperature	Humidity	Windspeed	Year	Count
x	10.66	56	11	2012	5600
x_{+j}	10.66	56	random	2012	4900
x_{-j}	10.66	56	random	random	

$\hat{f}$ $\Delta(j, S_m)$ marginal contribution

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SHAPLEY VALUE APPROXIMATION - ILLUSTRATION

Definition

average the contributions of feature j

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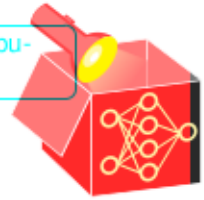

$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\mathbf{x}_{S_m \cup j}^{(m)}) - \hat{f}(\mathbf{x}_{S_m}^{(m)})\right] = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})\right]$$

- Compute marginal contribution of feature j towards the prediction across all randomly drawn feature coalitions $S_1, \ldots, S_m$
- Average all M marginal contributions of feature j
- Shapley value ϕ_j is the payout of feature j, i.e., how much feature *year* contributed to the overall prediction in bicycle counts of a specific observation $\mathbf{x}$

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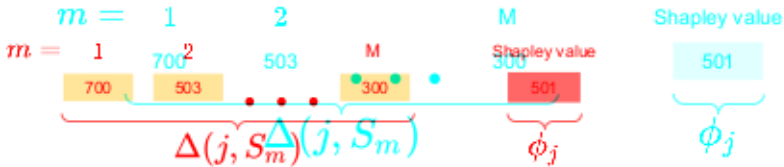

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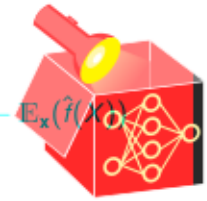
REVISITED: AXIOMS FOR FAIR ATTRIBUTIONS

We take the general axioms for Shapley Values and apply it to predictions:

- **Efficiency:** Shapley values add up to the (centered) prediction: $\sum_{j=1}^{p} \phi_j = \hat{f}(\mathbf{x}) - \mathbb{E}_{\mathbf{x}}(\hat{f}(X))$



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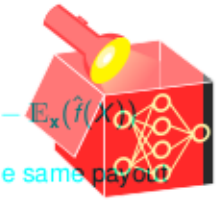

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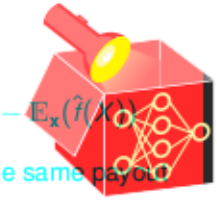
REVISITED: AXIOMS FOR FAIR ATTRIBUTIONS

We take the general axioms for Shapley Values and apply it to predictions:



- **Efficiency**: Shapley values add up to the (centered) prediction: $\sum_{j=1}^{p} \phi_j = \hat{f}(\mathbf{x}) - \mathbb{E}_{\mathbf{x}}(\hat{f}(X))$
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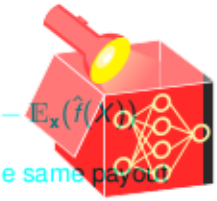

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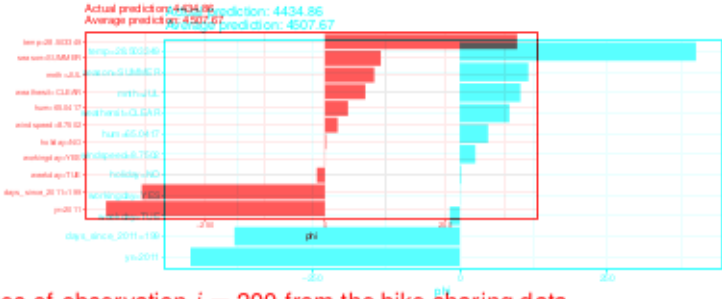
BIKE SHARING DATASET



- Shapley values of observation $i = 200$ from the bike sharing data
- Difference between model prediction of this observation and the average prediction of the data is fairly distributed among the features (i.e., $4434 - 4507 \approx -73$)
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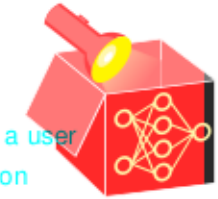
ADVANTAGES AND DISADVANTAGES



Advantages:

- **Solid theoretical foundation** in game theory
- Prediction is **fairly distributed** among the feature values — easy to interpret for a user
- **Contrastive explanations** that compare the prediction with the average prediction

Disadvantages:

- Without sampling, Shapley values need a lot of computing time to inspect all possible coalitions
- Like many other IML methods, Shapley values suffer from the inclusion of unrealistic data observations when features are correlated

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